

A Universal Square-Root-of-Six Bound for Kissing Numbers

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Abstract

Let τ_n denote the kissing number in Euclidean dimension n . We give an elementary proof that

$$\tau_n \leq (\sqrt{6})^n \quad (n \geq 1).$$

The constant $\sqrt{6}$ is optimal among bounds of the form $\tau_n \leq \alpha^n$ required to hold in every dimension, since $\tau_2 = 6$. The argument combines the elementary spherical-cap packing bound with a two-step recurrence for cap-volume fractions. An accompanying Lean 4/Mathlib development certifies the bound and its optimality by an alternative Euclidean-volume argument.

AI provenance and verification status. The proof and the first draft of this manuscript were generated by OpenAI’s GPT-5.6 Pro from a prompt supplied by the author. The Lean 4/Mathlib formalization, completed with further AI assistance in ChatGPT Work, proves the main theorem and its optimality without unproved geometric or analytic inputs. Its geometric argument differs from the cap-integral proof presented here; Section 3 records the distinction. The manuscript has not undergone independent human mathematical review.

1 Statement of the bound

For $n \geq 1$, let τ_n be the largest number of nonoverlapping unit balls in $\mathbb{R}^n$ that can simultaneously touch a fixed unit ball. Equivalently, τ_n is the largest cardinality of a subset of the unit sphere S^{n-1} whose pairwise angular distances are at least $\pi/3$.

The following answers the universal-base question that motivated this note.

Theorem 1. *For every integer $n \geq 1$,*

$$\tau_n \leq (\sqrt{6})^n.$$

Corollary 2. *The least real number $\alpha > 0$ such that*

$$\tau_n \leq \alpha^n \quad \text{for every } n \geq 1$$

is $\sqrt{6}$. Equivalently,

$$\sup_{n \geq 1} \tau_n^{1/n} = \sqrt{6}.$$

2 Proof

For $n \geq 2$, define

$$c_n := \frac{\int_0^{\pi/6} \sin^{n-2} t \, dt}{\int_0^{\pi} \sin^{n-2} t \, dt}. \tag{1}$$

This is the normalized surface measure of a spherical cap of angular radius $\pi/6$ in S^{n-1} . Indeed, polar coordinates about the center of the cap give the surface element as a constant multiple of $\sin^{n-2} t dt$, and the constant cancels in the ratio (1).

If the centers of two such caps have angular distance at least $\pi/3$, then their interiors are disjoint. The boundaries have surface measure zero, so every kissing configuration gives

$$\tau_n c_n \leq 1. \quad (2)$$

We next obtain a uniform lower bound for c_n .

Lemma 3. *For every $n \geq 2$,*

$$c_{n+2} > \frac{c_n}{6}.$$

Proof. Put $m = n - 2$ and set

$$I_m := \int_0^{\pi/6} \sin^m t dt, \quad J_m := \int_0^\pi \sin^m t dt,$$

so that $c_n = I_m/J_m$. Integration by parts gives

$$I_{m+2} = \frac{m+1}{m+2} I_m - \frac{\sqrt{3}}{(m+2)2^{m+2}}. \quad (3)$$

$$J_{m+2} = \frac{m+1}{m+2} J_m. \quad (4)$$

Consequently,

$$c_{n+2} = c_n - \frac{\sqrt{3}}{(m+1)2^{m+2}J_m}. \quad (5)$$

It remains to bound I_m from below. With the substitution $u = \sin t$,

$$I_m = \int_0^{1/2} \frac{u^m}{\sqrt{1-u^2}} du.$$

For $0 \leq u \leq 1/2$,

$$\frac{1}{\sqrt{1-u^2}} \geq 1 + \frac{u^2}{2}. \quad (6)$$

To see this, both sides are nonnegative and

$$(1-u^2) \left(1 + \frac{u^2}{2}\right)^2 = 1 - \frac{3u^4}{4} - \frac{u^6}{4} \leq 1.$$

It follows that

$$\begin{aligned} I_m &\geq \int_0^{1/2} u^m \left(1 + \frac{u^2}{2}\right) du \\ &= \frac{1}{2^m(m+1)} \left(\frac{1}{2} + \frac{m+1}{16(m+3)}\right) \\ &\geq \frac{25}{48 \cdot 2^m(m+1)}. \end{aligned}$$

Since $\sqrt{3} < 26/15$, we have

$$\frac{3\sqrt{3}}{10} < \frac{13}{25} < \frac{25}{48},$$

and hence

$$I_m > \frac{3\sqrt{3}}{10(m+1)2^m}. \quad (7)$$

Using $J_m > 0$, inequality (7) yields

$$\frac{\sqrt{3}}{(m+1)2^{m+2}J_m} < \frac{5}{6} \frac{I_m}{J_m} = \frac{5}{6} c_n.$$

Substitution into (5) proves $c_{n+2} > c_n/6$. □

Proof of Theorem 1. First suppose that n is even. Since $c_2 = 1/6$, Lemma 3 gives

$$c_n \geq 6^{-n/2}.$$

Together with (2), this implies

$$\tau_n \leq \frac{1}{c_n} \leq 6^{n/2} = (\sqrt{6})^n.$$

For odd $n \geq 5$, direct integration gives

$$c_5 = \frac{1}{2} - \frac{9\sqrt{3}}{32}.$$

The estimate $\sqrt{3} < 26/15$ implies $c_5 > 1/80$, whereas

$$(\sqrt{6})^5 = 36\sqrt{6} > 80$$

because $\sqrt{6} > 20/9$. Thus $c_5 > 6^{-5/2}$, and another application of Lemma 3 gives

$$c_n > 6^{-n/2} \quad \text{for every odd } n \geq 5.$$

Equation (2) again yields the desired bound.

It remains to consider $n = 1$ and $n = 3$. In dimension one, $\tau_1 = 2 < \sqrt{6}$. In dimension three,

$$c_3 = \frac{2 - \sqrt{3}}{4} > \frac{1}{15},$$

where the last inequality follows from $\sqrt{3} < 26/15$. Hence (2) gives $\tau_3 < 15$, so the integrality of τ_3 gives $\tau_3 \leq 14$. Finally,

$$14 < 6\sqrt{6} = (\sqrt{6})^3,$$

because $\sqrt{6} > 7/3$. This completes the proof. □

Proof of Corollary 2. Theorem 1 shows that $\alpha = \sqrt{6}$ works. Conversely, $\tau_2 = 6$: six equally spaced points on S^1 form a kissing configuration, while pairwise angular separation at least $\pi/3$ allows at most six points around the circle. Therefore every universal base α satisfies

$$6 = \tau_2 \leq \alpha^2,$$

and hence $\alpha \geq \sqrt{6}$. □

Remark 4. The argument is deliberately elementary and is not intended to compete asymptotically with the best high-dimensional spherical-code bounds. Its point is that $\sqrt{6}$ is the optimal constant in a dimension-uniform estimate having the particularly simple form $\tau_n \leq \alpha^n$ with no additional prefactor.

3 Scope of the Lean certificate

The accompanying repository is

<https://github.com/sebastienpalcoux/sqrt6-kissing-bound>.

It pins Lean 4.33.1 and Mathlib revision 0df444a360eaa60ab8c11dca51a86af692955474. The formal development defines a finite kissing configuration to be a set of unit vectors whose distinct members have inner product at most $1/2$. It proves, from exactly these hypotheses, that its cardinality is at most $(\sqrt{6})^n$ for every $n \geq 1$. The kissing number is then constructed as the greatest realizable cardinality, and both its realization and maximality are proved. An explicit regular hexagon establishes $\tau_2 = 6$ and the optimality of the universal base.

The formal proof uses disjoint measurable bodies inside the open 30° cones associated with the configuration. Writing V_n for the volume of the unit ball in $\mathbb{R}^n$, an inscribed bicone has volume

$$\frac{V_{n-1}}{n2^{n-1}}.$$

Fubini’s theorem and exact one-dimensional integration establish this formula. For $n \geq 6$, the inequalities $V_n \leq V_{n-1}$ and $n2^{n-1} \leq (\sqrt{6})^n$ finish the argument. Dimensions 1 through 5 are treated separately; the three- and four-dimensional cases use explicit profiles with exactly computed volumes. No exact three- or four-dimensional kissing number is assumed.

Thus the certificate proves Theorem 1 and Corollary 2 unconditionally, but it does not certify every intermediate identity of the cap-integral proof in Section 2. In particular, the general identity identifying the spherical-cap area with the sine-integral ratio (1) remains outside the formalization. The analytic recurrence and a chord-based lower estimate are verified separately as companion lemmas. The repository file `FORMALIZATION_SCOPE.md` describes the proof dependencies and the low-dimensional constructions.

The command `bash scripts/check.sh` builds the complete development, scans the project for proof holes and new axioms, and audits the axiom dependencies of all project declarations. Only the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound` are allowed. These checks establish formal validity relative to Lean and the pinned Mathlib development; they do not replace an independent human assessment of the manuscript.

Acknowledgment

This note was prompted by the MathOverflow question “Upper bound of the kissing number in n dimensions” and by Henry Cohn’s answer explaining how effective spherical-code bounds bear on the problem.

References

- [1] S. Palcoux, *Upper bound of the kissing number in n dimensions*, MathOverflow question 282644 (2017), <https://mathoverflow.net/questions/282644/>.
- [2] H. Cohn, *Answer to “Upper bound of the kissing number in n dimensions”*, MathOverflow answer 282685 (2017), <https://mathoverflow.net/a/282685/>.

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